

Marginal Distributions & Independence.

Theorem. Say that $(Y_1, \dots, Y_n)$ has the joint pdf $f_{Y_1, \dots, Y_n}(\dots)$.

Then, for every $i=1..n$, the random variable Y_i is also continuous w/ its marginal density

$$f_{Y_i}(y) = \underbrace{\int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty}}_{n-1} f_{Y_1, \dots, Y_n}(y_1, \dots, y_{i-1}, y, y_{i+1}, \dots, y_n) dy_n \dots dy_{i+1} dy_{i-1} \dots dy_1$$

Example. [cont'd]

(Y_1, Y_2) w/ joint pdf

$$f_{Y_1, Y_2}(y_1, y_2) = 6y_1 \mathbb{1}_{[0 \leq y_1 \leq y_2 \leq 1]}$$

Marginal of Y_1 = ?

$$\begin{aligned} f_{Y_1}(y) &= \int_{-\infty}^{\infty} f_{Y_1, Y_2}(y, y_2) dy_2 \\ &= \int_{-\infty}^{\infty} 6y \mathbb{1}_{[0 \leq y \leq y_2 \leq 1]} dy_2 \\ &= 6y \int_y^1 dy_2 = 6y(1-y) \mathbb{1}_{[0,1]}(y) \end{aligned}$$

Marginal of Y_2 = ?

$$\begin{aligned} f_{Y_2}(y) &= \int_{-\infty}^{\infty} 6y_1 \mathbb{1}_{[0 \leq y_1 \leq y \leq 1]} dy_1 \\ &= 6 \int_0^y y_1 dy_1 \mathbb{1}_{[0,1]}(y) = 6 \cdot \frac{y^2}{2} \mathbb{1}_{[0,1]}(y) = 3y^2 \mathbb{1}_{[0,1]}(y) \end{aligned}$$

Independence.

Theorem. The Factorization Criterion.

Jointly continuous r.v.s $(Y_1, \dots, Y_n)$ are
independent

iff

$$f_{Y_1, \dots, Y_n}(y_1, \dots, y_n) = f_{Y_1}(y_1) \cdot f_{Y_2}(y_2) \cdots f_{Y_n}(y_n) \text{ for all } y_1, \dots, y_n \in \mathbb{R}$$

Corollary. If

$$f_{Y_1, \dots, Y_n}(y_1, \dots, y_n) = g_1(y_1) \cdot g_2(y_2) \cdots g_n(y_n) \text{ for all } y_1, \dots, y_n$$

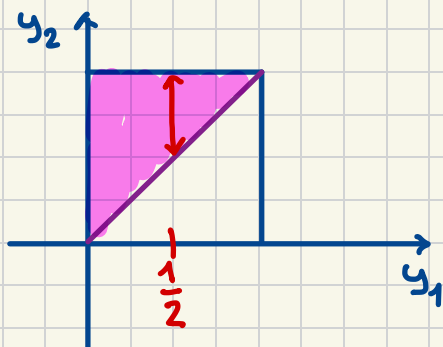
and some f'ctions $g_1, \dots, g_n$,

then, r.v.s $Y_1, \dots, Y_n$ are independent.

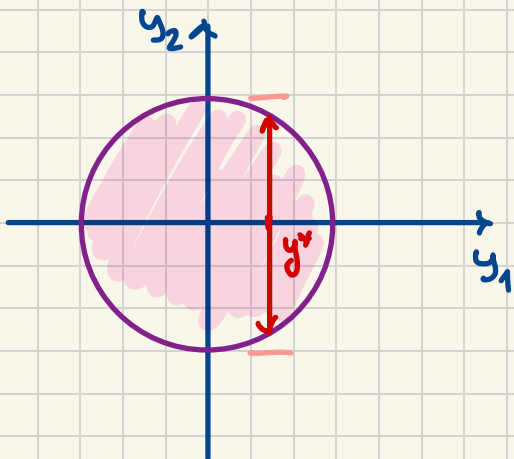
Example. [cont'd]

$$f_{Y_1, Y_2}(y_1, y_2) \stackrel{?}{=} f_{Y_1}(y_1) \cdot f_{Y_2}(y_2)$$

$$6y_1 \mathbb{1}_{[0 \leq y_1 \leq y_2 \leq 1]} \neq 6y_1(1-y_1) \mathbb{1}_{[0,1]}(y_1) \cdot 3y_2^2 \mathbb{1}_{[0,1]}(y_2)$$



Example.



Theorem. Let $Y_1, \dots, Y_n$ be independent r.v.s,
and let $h_1, \dots, h_n$ be functions such that
 $h_i(Y_i), i=1..n$
are all well-defined.

Then, if all expectations are finite, we have

$$\mathbb{E}[h_1(Y_1) \cdot h_2(Y_2) \cdots h_n(Y_n)] = \mathbb{E}[h_1(Y_1)] \cdot \cdots \cdot \mathbb{E}[h_n(Y_n)]$$

e.g., Y_1, Y_2 independent

$$h_1(y) = h_2(y) = e^y \text{ for all } y \in \mathbb{R}$$

$$\mathbb{E}[\exp(Y_1 + Y_2)] = \mathbb{E}[e^{Y_1} \cdot e^{Y_2}] \stackrel{\text{independence}}{=} \mathbb{E}[e^{Y_1}] \cdot \mathbb{E}[e^{Y_2}]$$

If Y_1 and Y_2 are identically dist'd, i.e., if $F_{Y_1} = F_{Y_2}$,
then,

$$\mathbb{E}[\exp(Y_1 + Y_2)] = (\mathbb{E}[e^{Y_1}])^2.$$

M378K Introduction to Mathematical Statistics

Problem Set #8

Transformations of Random Variables.

Problem 8.1. Let X be a continuous random variable with the cumulative distribution function denoted by F_X and the probability density function denoted by f_X .

Let the random variable $Y = 2X$ have the p.d.f. denoted by f_Y . Then,

(a) $f_Y(x) = 2f_X(2x)$

(b) $f_Y(x) = \frac{1}{2}f_X\left(\frac{x}{2}\right)$

(c) $f_Y(x) = f_X(2x)$

(d) $f_Y(x) = f_X\left(\frac{x}{2}\right)$

(e) None of the above

The CDF Method

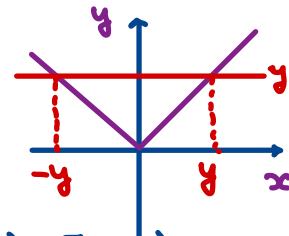
$$\begin{aligned} \rightarrow: y \in \mathbb{R} : F_Y(y) &= \mathbb{P}[Y \leq y] = \mathbb{P}[2X \leq y] \\ &= \mathbb{P}\left[X \leq \frac{y}{2}\right] = F_X\left(\frac{y}{2}\right) \end{aligned}$$

$$f_Y(y) = \frac{d}{dy} F_X\left(\frac{y}{2}\right) = \frac{1}{2} f_X\left(\frac{y}{2}\right)$$

□

Problem 8.2. If the continuous random variable X has the distribution function F_X , then the distribution function of the random variable $Y = |X|$ equals

$$\begin{aligned} \rightarrow: & \text{for } y < 0 : F_Y(y) = \mathbb{P}[Y \leq y] = \mathbb{P}[Y < 0] = 0 \\ & \text{for } y \geq 0 : F_Y(y) = \mathbb{P}[Y \leq y] = \mathbb{P}[|X| \leq y] \\ & = \mathbb{P}[-y \leq X \leq y] = \\ & = \mathbb{P}[X \leq y] - \mathbb{P}[X < -y] = F_X(y) - F_X(-y) \end{aligned}$$



For fun: $f_Y(y) = (f_X(y) + f_X(-y)) \mathbb{1}_{[0, \infty)}(y)$

Remark 8.1. The goal is to figure out the distribution of the random variable

$$X = g(Y_1, Y_2, \dots, Y_n)$$

where $Y_i, i = 1, \dots, n$ are a random sample with a common density f_Y .

Def'n. $Y_1, \dots, Y_n$ is a random sample from a dist'n $\mathcal{D}$ if:

- $Y_1, \dots, Y_n$ are independent
- and
- $Y_i \sim \mathcal{D}$ for all $i=1..n$